



Does The Heart Gate Perception?

Behavioral evidence from cardiac-phase-triggered visual tasks.

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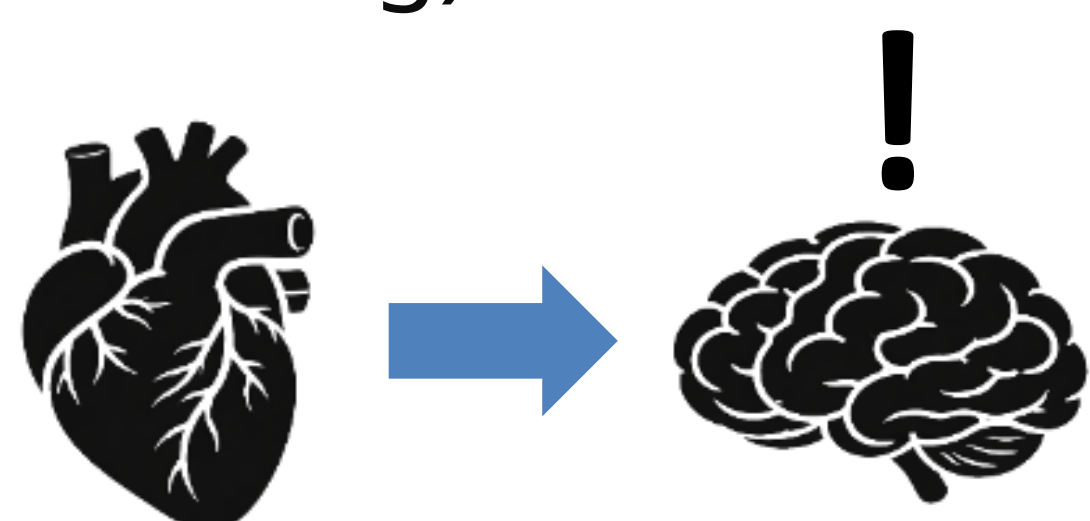


Introduction

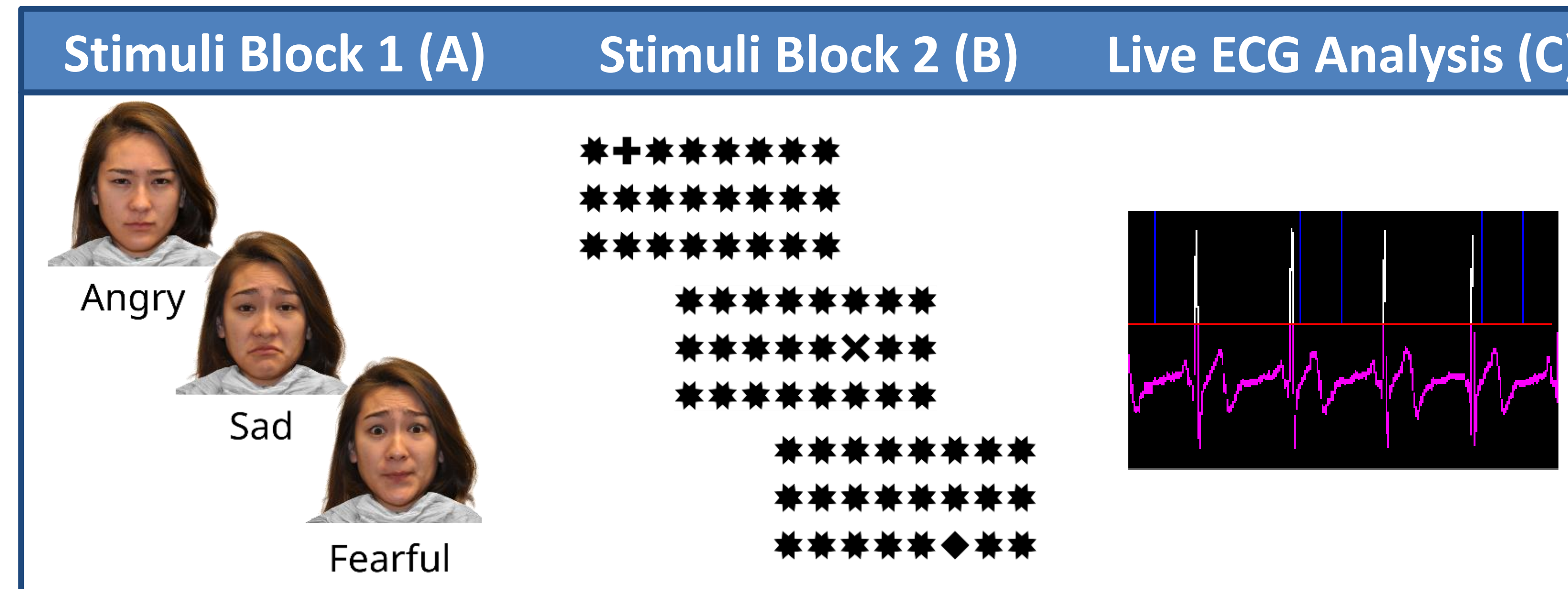
Perception is not constant—our experience of the world is influenced by what is happening inside our bodies. Interoceptive signals, like those following a heartbeat, can influence ongoing neuronal excitability across cortical structures. During **systole** (when the heart contracts), sensory processing may be reduced, while during **diastole** (when the heart relaxes), perception may be enhanced. This is reflected in an altered perception or encoding of somatosensory stimuli, such as pain, touch, auditory input, and even the experience of time. However, these influences are variable across sensory modalities and across different stages of sensory processing. In this study, we test whether the timing of a visual stimulus within the heartbeat affects **objective perception**, using very brief presentations to focus on early visual processing.

Hypothesis

Visual accuracy will be higher during diastole (minimal baroreceptor firing) than systole (maximal baroreceptor firing).



Methods



Design: Within-subject design

Participants:

- $N \approx 48$ college students

Stimuli:

- Emotional faces (A)
- Symbol detection task (B)

Procedure:

- 16 ms presentation (near-threshold)
- Systole (~ 100 & 200 ms post-R) (C)
- Diastole (~ 500 ms post-R) (C)

Try it yourself!

Scan to see the stimulus animation



Discussion

We found no evidence that perception is better during diastole, despite theoretical predictions. Accuracy did not differ across cardiac phases. However, reaction time was affected, suggesting that the heartbeat influences how we process information **after** perception rather than during early visual processing.

These findings indicate that internal bodily signals, like the heartbeat, may shape later stages of thinking and interpretation rather than what we initially see. This may help us better understand conditions such as anxiety, where sensitivity to bodily signals is altered.

Future research will explore how heart-brain interactions influence perception and cognition using brain imaging and clinical populations.

Results

Identification Accuracy for Near-Threshold Faces and Symbols During Systole and Diastole

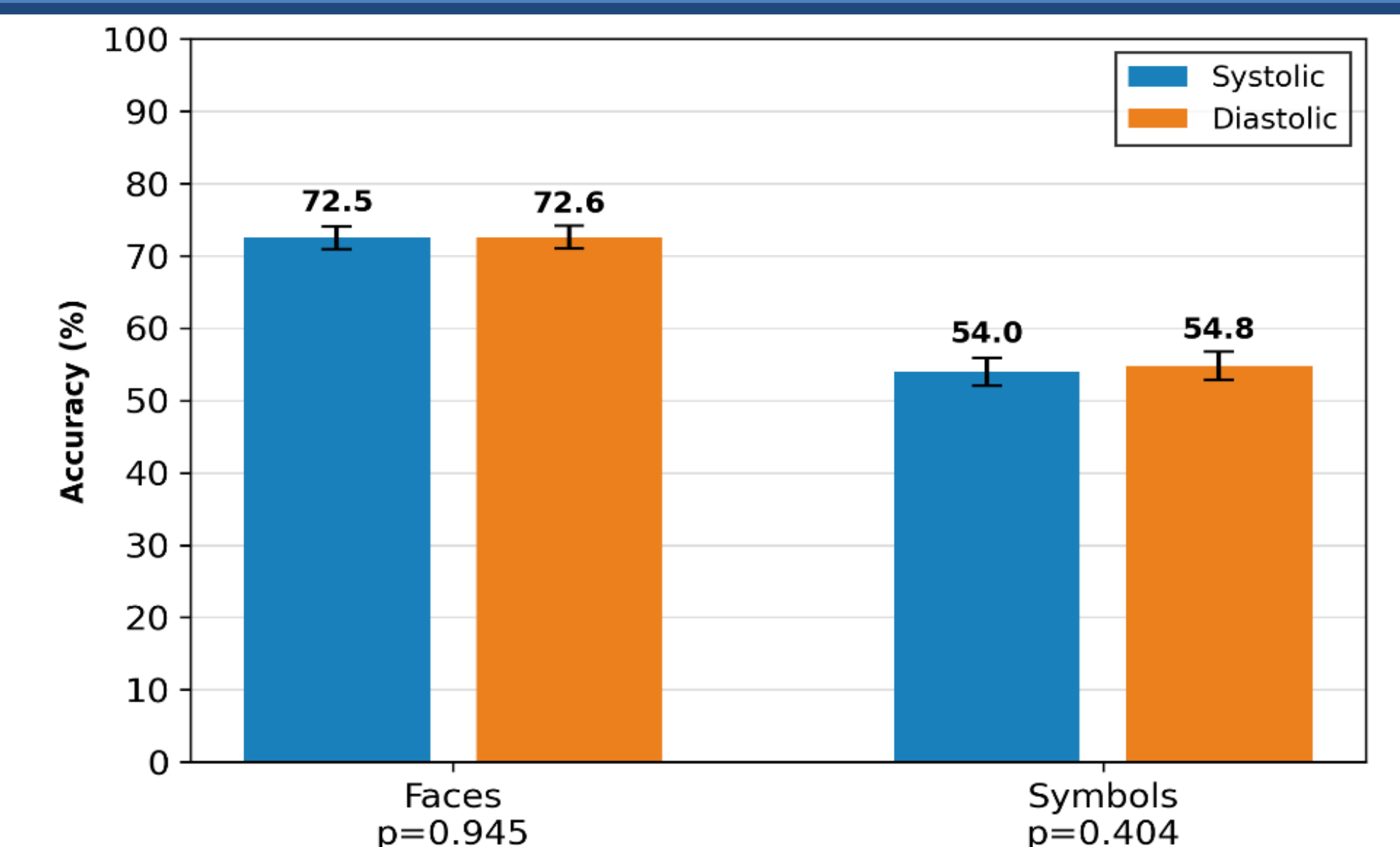


Figure 1. : Accuracy did not differ between systolic and diastolic presentations for either the emotional face or symbolic detection tasks, providing no evidence for a diastolic advantage in perceptual access.

Response Times for Near-Threshold Faces and Symbols During Systole and Diastole

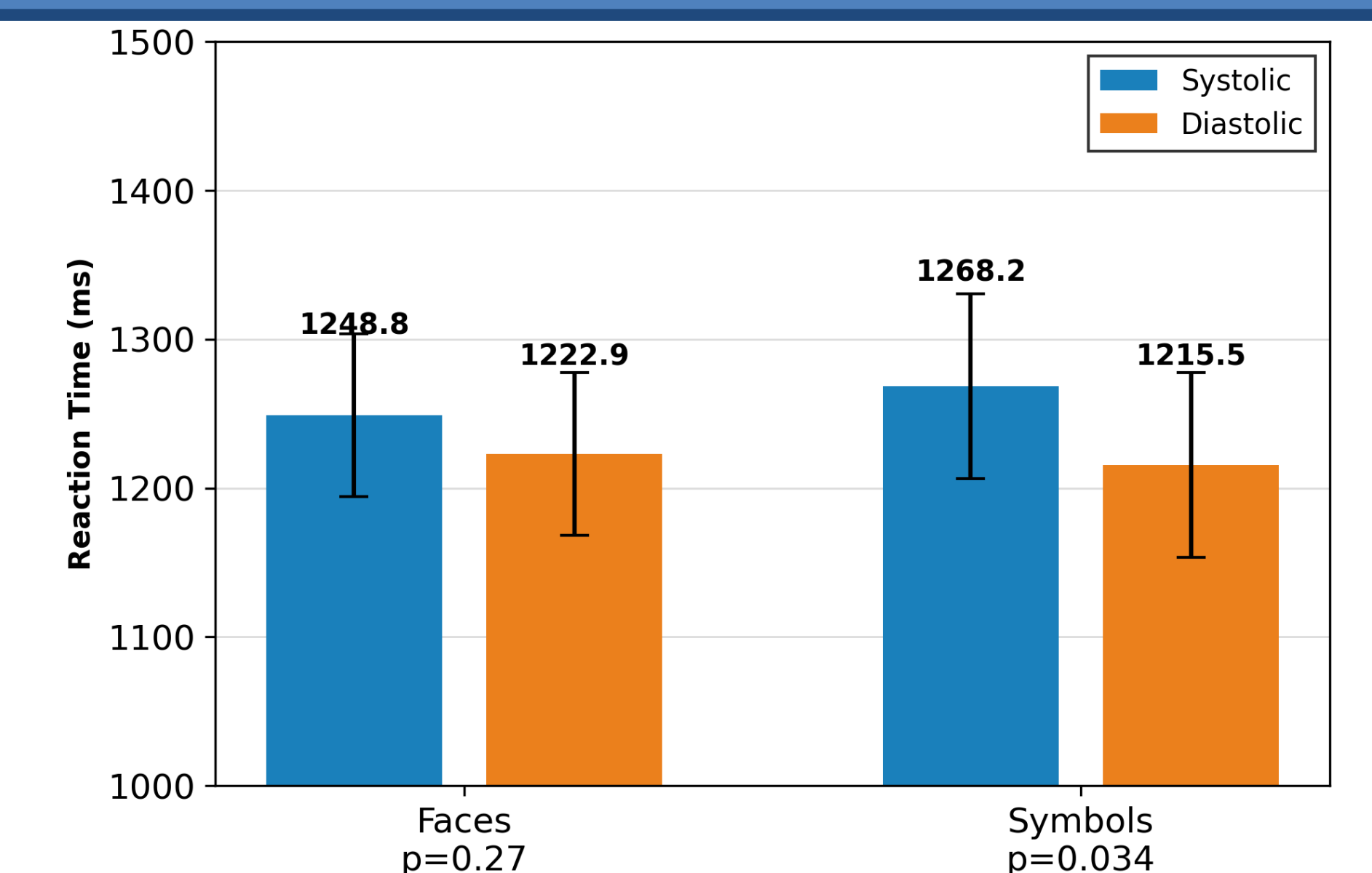


Figure 2. : Reaction times were slower during systole compared to diastole for symbolic stimuli ($p \approx .03$), with no significant differences for faces, indicating that cardiac phase effects may emerge at the level of response processing rather than perception.

Acknowledgements

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